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Multiple hoist delivery system for delivering parcels using unmanned aerial vehicles

Abstract

A multiple hoist system is used with an unmanned aerial vehicle (UAV) for delivering parcels. An example of the multiple hoist system comprises two or more hoists that are independently operable, meaning that a first hoist can lower or raise a first line independently of using a second hoist to raise or lower a second line. The hoists can independently raise and lower their associated lines to allow the UAV to deliver multiple parcels to multiple delivery locations, or the hoists can synchronously raise and lower the associated lines together so that larger parcels can be delivered using the UAV. The hoists can be comprised within a body of the multiple hoist system. The body can further include a securing device for releasably securing the multiple hoist system to the UAV.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/108,613, filed on Dec. 1, 2020, and entitled “Multiple Hoist Delivery System for Delivering Parcels Using Unmanned Aerial Vehicles,” which claims the benefit of priority to U.S. Provisional Patent Application No. 62/942,493, filed on Dec. 2, 2019, and entitled “Multiple Hoist Delivery System for Delivering Parcels Using Unmanned Aerial Vehicles,” each of which is expressly incorporated by reference in its entirety.

BACKGROUND

(1) Delivering parcels using drones, in particular unmanned aerial vehicles (UAVs), is becoming a reality. Currently, there are some UAVs that deliver parcels by lowering the parcel downward from the UAV using a line. Lowering parcels downward from an altitude of the UAV is beneficial because it allows the UAV to remain at a safe distance above a surface, so that it can avoid interaction with objects and people. This increases the safety of UAV delivery, and it allows UAV delivery to a greater number of locations, such as locations that would be difficult for a UAV to traverse if it were to deliver a parcel close to the surface.

(2) The drawback, however, is that current systems have only one line. This restricts these systems to delivering a single parcel or one group of parcels to a single location. Once the parcels are released from the line, the UAV must navigate back to a loading point to pick up another parcel and continue delivery.

SUMMARY

(3) At a high level, aspects described herein relate to a UAV having a multiple hoist system. The multiple hoist system comprises at least two hoists that are independently operable. By having independently operable hoists, a first line associated with a first hoist can be lowered independently of a second line associated with a second hoist. This allows for multiple parcels to be loaded onto the UAV and delivered to multiple delivery locations. This also allows for a single, larger parcel to be secured at multiple points using both the first and second lines. The lines can be synchronously lowered or raised to deliver the larger parcel. In this way, the multiple hoist system provides for a more versatile UAV delivery system.

(4) Another aspect of the multiple hoist system includes a body that houses the at least two hoists. The body can include a securing device, such as a securing hook, latch, threaded system, etc. that releasably engages a securing area of the UAV. In this manner, the multiple hoist system can be easily removed from and attached to a UAV.

(5) This summary is intended to introduce a selection of concepts in a simplified form that is further described in the Detailed Description section of this disclosure. The Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. Additional objects, advantages, and novel features of the technology will be set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the disclosure or learned through practice of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology is described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1 is an example operating environment employing a multiple hoist system with a UAV, in accordance with an aspect described herein;

(3) FIG. 2 is a side view of an example multiple hoist system affixed to a UAV, in accordance with an aspect described herein;

(4) FIG. 3 is a side view of another example multiple hoist system having a side surface of a body removed for viewing within the body, in accordance with an aspect described herein;

(5) FIG. 4 is a side perspective view of another example multiple hoist system, in accordance with an aspect described herein;

(6) FIG. 5 is an illustration of a UAV engaging in independent operation of a first and second hoist of an example multiple hoist system, in accordance with an aspect described herein;

(7) FIG. 6 is a side view of an example hoist having a line in use with curved contact points, in accordance with an aspect described herein;

(8) FIG. 7 is an illustration of a UAV engaging in synchronous operation of a first and second hoist of an example multiple hoist system, in accordance with an aspect described herein;

(9) FIG. 8 is a block diagram illustrating an example method of delivering parcels using a multiple hoist system;

(10) FIG. 9 is a block diagram of an example computing device suitable for use with aspects of the present technology; and

(11) FIG. 10 is an illustration of a UAV with a single reel and multiple lines for synchronous raising and lowering of a parcel, in accordance with an aspect described herein.

DETAILED DESCRIPTION

(12) FIG. 1 depicts a block diagram of example operating environment **100** for UAV delivery using a multiple hoist delivery system. Illustrated in the example operating environment **100** is UAV **102** having an attached multiple hoist delivery system **104**. UAV **102** is in communication via network **106** to on-board computing device **108**. It should be understood that operating environment **100** shown in FIG. 1 is an example of one suitable operating environment, and that other arrangements, including more or less components, are also suitable.

(13) UAV **102** can be any type of UAV. In general, a UAV includes any flight-capable vehicle that can be controlled remotely by a human pilot, operate autonomously based on a set of received or determined instructions, or a combination of both. UAVs may include vertical take-off and landing (VTOL) aircraft, which are aircraft that can hover, take off, or land vertically or near vertically. Examples of these include single-rotor VTOL aircraft, such as helicopters, or multi-rotor VTOL

aircraft. UAV is also intended to include conventional take-off and landing (CTOL) aircraft. CTOL aircraft generally take off and land over a distance. Many CTOLs include fixed-wing aircraft that use an airfoil for forward flight advantages. UAV is also intended to include aircraft using a combination of rotor(s) and a fixed wing. Thus, while UAV **102** is illustrated as a fixed-wing CTOL aircraft, in practice, it may be any type of UAV, including a multi-rotor VTOL aircraft.

(14) As illustrated, UAV **102** has attached multiple hoist system **104**. Multiple hoist system **104** is configured to be releasably secured to UAV **102**. Multiple hoist system **104** includes a plurality of hoists that are independently movable. In this way, one or more parcels may be lowered and raised by UAV **102** from safe altitudes. Various embodiments of multiple hoist systems that are suitable for use as multiple hoist system **104** will be described in more detail throughout this disclosure.

(15) With continued reference to FIG. **1**, UAV **102** communicates via network **106** to on-board computing device **108**. Network **106** encompasses any form of wired or wireless communication. Wireless communication examples include one or more networks, such as a public network or virtual private network “VPN.” Network **106** may include one or more local area networks (LANs) wide area networks (WANs), or any other communication network or method. In addition to Wi-Fi, other wireless examples include Bluetooth and infrared communication methods.

(16) UAV **102** comprises on-board computing device **108**. An example of on-board computing device **108** includes a flight controller. Various flight controllers are available for use with UAV **102**. One of ordinary skill in the art will have an understanding of the availability and benefits of such flight controllers. Though represented as a single component, on-board computing device **108** can be distributed in nature. That is, one or more functions may be performed by a single component or by a plurality of components distributed throughout UAV **102**. On-board computing device **108** generally includes a processor that executes instructions stored on computer memory. An example includes computing device **900** of FIG. **900**.

(17) On-board computing device **108** may receive instructions from remote server **110**. To do so, on-board computing device **108** can be in communication with a receiver (not illustrated) to receive signal **112** from remote server **110**. On-board computing device **108** may also communicate to remote server **110** using transmitter **114** that sends signal **116**.

(18) In general, receivers and transmitters comprise any system known in the art for sending and receiving communication signals, such as those represented by signal **112** and signal **116**. Examples may include receivers and transmitters that send and receive signals across a frequency band of the electromagnetic spectrum. Some suitable short-distance transmitter-receiver pairs are configured for use with 900 MHz, 2.4 GHz, and 5.8 GHz communication bands. Some over-the-horizon (OTH) transmitter-receiver pairs utilize telecommunications bands, such as those supporting LTE, 4G, and 5G bands. Other example transmitter-receiver pairs use frequency bands associated with satellite communications. These are just examples, and other suitable communication methods may be used and are contemplated to be within the scope of this disclosure.

(19) With reference back to FIG. **1**, it should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, and groupings of functions, etc.) can be used in addition to or instead of those shown, and some elements may be omitted altogether for the sake of clarity. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by one or more entities may be carried out by hardware, firmware, or software. For instance, some functions may be carried out by a processor executing instructions stored in memory as further described with reference to FIG. **9**.

(20) Turning now to FIG. **2**, a side view of example multiple hoist system **202** is illustrated. As illustrated, multiple hoist system **202** is secured to UAV **200**. Multiple hoist system **202** comprises body **206**. Body **206** houses at least two hoists, including first hoist **208** having first line **209** and second hoist **210** having second line **211**.

(21) As shown, multiple hoist system **202** is secured to UAV **200** using securing hooks **204A** and **204B**. While there are two securing hooks illustrated in FIG. 2, any number of securing hooks, including a single securing hook may be used. Here, securing hooks **204A** and **204B** are illustrated as affixed to body **206** of multiple hoist system **202**. Securing hooks **204A** and **204B** can be affixed to an upper surface of body **206**. In another example, securing hooks **204A** and **204B** may be affixed to UAV **200**.

(22) In general, each securing hook can have a corresponding securing area for which to secure. Securing areas can include a recessed area or opening shaped to receive securing hooks, or any other area of contact suitable for releasably securing hooks, or a securing device, more generally. As illustrated, where securing hooks **204A** and **204B** are affixed to body **206**, the corresponding securing areas are found on UAV **200**. In an embodiment not illustrated, where securing hooks (or a securing device, more generally) are affixed to a UAV, the securing areas may be located on the multiple hoist system body.

(23) Securing hooks **204A** and **204B** are suitable for releasably securing multiple hoist system **202** to UAV **200**. To do so, securing hooks **204A** and **204B** can be placed within corresponding securing areas on UAV **200**. Securing hooks **204A** and **204B** can then be moved to a securing position to releasably secure multiple hoist system **202**.

(24) It will be appreciated that other securing devices and methods for releasably securing multiple hoist system **202** to UAV **200** can be used. The use of securing hooks, generally, is one suitable method that can be used with the present technology. However, others are contemplated and are intended to be within the scope of this disclosure. Among many other examples of securing devices, the use of locks and pins, threaded systems, and latches may be used in lieu of or in addition to the securing methods illustrated and described herein.

(25) Body **206** of multiple hoist system **202** houses first hoist **208** and second hoist **210**. As illustrated, a portion of first hoist **208** and second hoist **210** extends outward from a bottom surface of body **206**. Thus, in this example, body **206** houses only a portion of first hoist **208** and second hoist **210**, while another portion of first hoist **208** and second hoist **210** is external to body **206**. As will be described in more detail with other examples, the body generally comprises two or more hoists, but various suitable arrangements of the hoists within the body may be used. For example, hoists may be fully housed within the body.

(26) Hoists, such as first hoist **208** and second hoist **210**, are generally configured to lower a load using a line, such as first line **209** and second line **211**, respectively. In addition to lowering a load, some hoists can also be configured to raise a load. A load comprises any weighted object, including the line itself. The load may comprise a parcel being delivered to a location by a carrier operating UAV **202**.

(27) In general, a number of hoists are suitable for use with the present technology. One consideration in selecting a hoist is the weight of the hoists versus the amount of load the hoists can raise or lower. When using larger UAVs that can carry parcels over long distances and have greater payload capacity, relatively heavier hoists may be selected for use. This provides the advantage of being able to raise and lower heavier parcels. Alternatively, where smaller UAVs are utilized for shorter delivery distances and that have lower payload capacities, relatively smaller hoists may be selected for use. This provides the advantage of weight reduction, allowing for more of the UAVs payload capacity to be dedicated to the parcel being delivered.

(28) Some example hoists that can be used are commercial hoists, such as electric overhead hoists. These hoists are generally powered by an electric motor, which can receive a power supply from a power source associated with multiple hoist system **202** or with UAV **200**. These commercial hoists are available in various weights and have various load capacities. In some embodiments, example hoists can include a multiple tether system linked to a single motor that is able to selectively actuate each of a plurality of tether reels for raising and lower the lines. For example, a winch with a single motor and a gearbox type mechanism can engage several tether reels lined up on a shared

axle, lowering or reeling up each of the lines independently. The traction on each reel can be accomplished with an in-hub system, similar to those used on some bicycles, thus reducing the need to have more than one motor and resulting in a reduction in weight for the hoists. However, other hoists can be used without departing from the scope of the technology described herein. (29) Since weight is generally a consideration in the selection of hoists, another suitable device that can be used is an electric filament reel. Commercial monofilaments are available that can bear heavy loads, yet are lightweight. Among others, some examples include nylon, polyvinylidene fluoride (PVDF), polyethylene, polyethylene terephthalate (PET), and ultra-high-molecular-weight polyethylene (UHMWPE). Other biodegradable-type filaments can be selected for environmentally friendly and safe delivery. It should be understood, that unless expressly stated otherwise, the use of “hoist” and “electric reel” within this disclosure are synonymous. That is, where it is stated an electric reel may be used, it will be understood that a hoist could be used in addition to or alternatively, and vice versa.

(30) Reels configured to utilize filament-type line are generally lighter in weight. Electric motors operable with such reels are available having different torque specifications. Similar to commercial hoists, the electric motors having greater torque typically have a relatively greater weight and size. The load that can be handled using an electric motor and reel combination (also referred to as an “electric reel”) directly relates to the torque capacity of the motor. Thus, there can be a tradeoff between the torque specifications of an electric motor operating a reel. In this way, for UAVs designed to carry lighter loads, a relatively smaller and lighter weight electric motor and reel combination can be selected.

(31) To provide some non-exclusive examples, an electric motor and reel combination can be selected having a torque capacity of up to 5 lbs, up to 10 lbs, up to 15 lbs, up to 20 lbs, and so forth. Any of these may be used in conjunction with, for example, a monofilament line having a rating of up to 5 lbs, up to 10 lbs, up to 15 lbs, up to 20 lbs, and so forth.

(32) Reels may include line levelers, such as first line leveler **213** and second line leveler **214**, illustrated in FIG. 2. Line levelers are optional and may be provided to level the line placement along the reel when retracting a line. Thus, when retracting lines from higher altitudes, line levelers can wind lines more evenly along a reel to keep the line from accumulating in one place and binding during the next extension of the line. Line levelers that are configured to move parallel to the reel as the reel is turned can be utilized. Using line levelers may provide the added benefit of lowering loads, including parcels for delivery, from higher altitudes, thus increasing safety of UAV delivery and allowing a UAV access to a greater number of locations, such as those where low altitude deliveries are difficult. While the use of line levelers is optional, a line leveler may be employed where UAVs deliver parcels using the line from an altitude of up to 25 ft, up to 50 ft, up to 75 ft, up to 100 ft, and so forth.

(33) Lines can be wrapped around the body or the UAV to reduce direct force applied to the motor of the reel. With brief reference to FIG. 6, the figure illustrates an example hoist **600** (e.g., an electric reel) employed for use with line **602**. One or more curved contacts, such as curved contacts **604A** and **604B**, can be employed to modify the force applied to electric reel **600**. In this way, the force applied along line **602** by an attached load can be distributed to other points of a multiple hoist system or a UAV. For example, a curved contact point can be secured to a body of the multiple hoist system. Curved contact points can be secured within the body or may be secured to an external surface of the body. In another example, curved contact points are secured to an external surface of the UAV. In this way, a line extends from an electric reel to the curved contact points and downward in order to lower and raise loads.

(34) In some embodiments, simple tension sensors can be used to indicate when the parcel is released from the line. In some example embodiments, the arm where the curved contact **604B** is held can be spring loaded and a pressure sensor can be attached thereto. Specifically, a spring-loaded small pulley that pushes against the line can result in tension in the line or lack thereof, such

as if the parcel is resting on the ground or has been disconnected from the line. This lack of tension sensed by the tension sensors can indicate to the flight controller or any of the controllers described herein when the parcels attached to the line should be released (e.g., the parcel is resting on the ground) and when the corresponding hoist should raise the line back up (e.g., the parcel is detached therefrom).

(35) With reference back to FIG. 2, some aspects of multiple hoist system **202** comprise a line cutter (not illustrated). Line cutters include any device for severing a line, such as first line **209** and second line **211**. These may include systems having a sharp edge that is pressed against the line upon command from a flight controller or remote server. Other devices may apply heat or lasers to sever a line, and can also be controlled based on instructions received from the flight controller or remote server. Line cutters may be used to sever lines in the event of a line bind or snag.

(36) An example method for severing a line using a line cutter includes extending the line from the UAV. When the line is at an extended position, detecting a line overload. The line overload occurs where the force applied to the line exceeds a safety threshold. The safety threshold can be based on the torque capacity of the electric reel or the line weight rating. The flight controller can detect that the safety threshold has been exceeded based on an amount of thrust necessary to maintain a target altitude. For example, the greater the downward force being applied to the line, the greater the upward force applied by the UAV motors must be to maintain the target altitude. If the force applied to the line increases, the flight controller will increase the upward thrust from the UAV motors to counteract the increased force. When the upward thrust exceeds the safety threshold, the flight controller activates the line cutter and severs the line. This allows the extended line to drop from the UAV and allows the UAV to continue navigation away from the area where it severed the line. This increases the safety of UAV delivery by allowing the UAV to continue safe navigation in the event a malfunction with the line occurs. For instance, should a line be lowered from a UAV and catch on an object, such as a tree, then line cutter can sever the line and allow the UAV to continue flight. Further, should a person accidentally or maliciously pull an extended line during UAV parcel delivery, the UAV can sever the line and navigate away, as opposed to this causing a downed UAV.

(37) Continuing with FIG. 2, first hoist **208** and second hoist **210** are independently operable. That is first hoist **208** can extend or retract line **209** independent of using second hoist **210**. In this way, line **209** can be extended or retracted without regard to the position of line **211**. Similarly, second hoist **210** is operable independent of first hoist **208**, in that second hoist **210** can extend or retract line **211** without regard to the position of line **209**. Where more than two hoists are provided in a multiple hoist system, each hoist can be independently operated.

(38) As an example, hoists can generally be operable under guidance of the flight controller. The flight controller and the hoist can be in wired or wireless communication. In the example provided by FIG. 2, wired communication channel **212** is illustrated as providing a connection between components of UAV **200** and components of multiple hoist system **202**, such as providing a communication channel between a flight controller of UAV **200**, and first hoist **208** and second hoist **210**. Thus, the flight controller can independently operate each hoist.

(39) An illustration of independently operable hoists is provided by FIG. 5. In this example, UAV **500** is shown as a VTOL six-rotor multicopter. Secured to a bottom of UAV **500** is multiple hoist system **502**, which comprises first hoist **504** and second hoist **506**. First hoist **504** is configured to raise and lower line **505**. Attached to first line **505** is first parcel **508**. Second hoist **506** is configured to raise and lower second parcel **510** using a second line, which is not illustrated because of the raised position of parcel **510**.

(40) Because each of first hoist **504** and second hoist **506** are independently operable, parcels that are associated with each of the hoists can be independently lowered or raised. As illustrated in FIG. 5, parcel **508** can be lowered in the direction of first arrow **512** using first hoist **504**. Similarly, parcel **508** can be raised in the direction of second arrow **514** using first hoist **504**. In each of these

events, whether first parcel **508** is being raised or lowered, second parcel **510** can remain in a constant position, shown in FIG. 5 as a raise position adjacent a lower surface of multiple hoist system **502**. Similarly, independent of the movement of parcel **508**, parcel **510** can be raised or lowered by hoist **506**. In this way, UAVs can be configured to deliver multiple parcels to different delivery locations. That is, parcel **508** can be lowered to and released at a first delivery location, while second parcel **510** can be lowered to and released at a second delivery location. This allows for multiple deliveries in between loading or unloading of parcels onto the UAV.

(41) While hoists are generally independently operable, hoists can also be instructed to raise or lower lines synchronously. An example of this is illustrated in FIG. 7. FIG. 7 illustrates UAV **700**, which is represented as a VTOL six-rotor multicopter. Multiple hoist system **702** is secured to UAV **700**. Here, an upper surface of multiple hoist system **702** is secured to a lower surface of UAV **700**, which as previously described, the multiple hoist system **702** could be releasably secured or permanently secured to UAV **700**. Multiple hoist system **702** comprises first hoist **704** that operates to raise and lower first line **705** and second hoist **706** that operates to raise and lower second line **707**.

(42) Here, first hoist **704** and second hoist **706** are instructed to raise or lower first line **705** and second line **707**, respectively, in a synchronous manner. That is, synchronously raising or lowering first line **705** and second line **707** comprises instructing first hoist **704** to raise or lower first line **705** at the same rate as second hoist **706** is instructed to raise or lower second line **707**. In this manner, UAVs are more versatile, in that they can carry various size and shape parcels. As illustrated in FIG. 7, UAV **700** can lower parcel **708** in the direction of first arrow **710** or raise parcel **708** in the direction of second arrow **712** using two attachment points, where each attachment point to parcel **708** is associated with one of first line **705** or second line **707**.

(43) In another alternative embodiment, one or more of the hoists can also be instructed to raise or lower lines synchronously using multiple lines in a single reel of a single hoist. Specifically, the single reel can use line levelers (as earlier described herein) and each line can be threaded through a different curved contact or small pulley to spread them apart. This configuration can be used for one or more of the hoists, such that each hoist is connected to multiple lines. In one example embodiment, as depicted in FIG. 10, a UAV **1000** can have four lines **1010** or tethers can be operated by a single reel **1020** and winch in this manner, with the four lines **1010** separated via a plurality of curved contacts **1030**. This advantageously allows each line to be lighter, since the lines cooperatively share the load. This can also ensure that the load or parcel is lowered without spinning, which allows raising and securing the load or parcel in a correct position/orientation. In yet another embodiment, a single motor can similarly and synchronously operate a plurality of reels.

(44) With reference back to FIG. 2, hoists, such as first hoist **208** and second hoist **210** can include a line release and a line counter. The line release and the line counter can be either electronic or mechanical. A line release is operable to release the line from electronic control of the hoist. In this way, a force applied to the line will allow the line to extend from the hoist. For example, a hoist may have a load attached to an associated line. By releasing the line from electronic control of the hoist, the line begins to extend away from the hoist due to the force applied to the line by the load. Thus, for example, a parcel may be lowered from an altitude by releasing the hoist based on the force of gravity applied to the line because of the attached parcel. As such, a UAV carrying a parcel can lower the parcel downward to a delivery location based on gravitational force, rather than expend power to electronically lower the parcel. A line counter generally works to apply an opposing force to the line. It may further reestablish electronic control by the hoist.

(45) Using the line release and the line counter, a parcel may be lowered from an altitude above a delivery location by a UAV. The parcel can be initially be lowered by releasing the line using the line release and allowing gravity to lower the parcel. The line counter may engage at a predetermined rate of extension by the line. That is, to avoid continued constant acceleration and

allowing the parcel to descend too rapidly, the line counter can be applied to reduce downward acceleration to up to zero. At a predetermined altitude of the parcel above the delivery location, the line counter can begin to slow the parcel down. The predetermined altitude can be any altitude, including as examples, 3 ft, 5 ft, 10 ft, 15 ft, and so forth. The line counter can further reestablish electronic control by the hoist. Using this method, once the parcel is lowered below the predetermined parcel altitude, the hoist can then engage in electronic control of the line's extension. The hoist can then electronically control the rate at which the parcel descends downward to the delivery location. By doing so, the parcel can be placed on the ground as its downward velocity is reduced to approach zero under electronic control of the hoist. As such, the parcel can gently be placed onto a surface and released.

(46) Turning now to FIG. 3, a side view of example multiple hoist system **300** is provided. Multiple hoist system **300** comprises body **302**. Here, a side surface of body **302** is removed to view within body **302**. An upper surface of body **302** is illustrated as having securing hooks **304A** and **304B**, which can be used for releasably securing multiple hoist system **300** to a UAV, not illustrated in FIG. 3.

(47) As provided by FIG. 3, multiple hoist system **300** comprises two hoists, first hoist **306** and second hoist **308**. Each of the hoists is at least partially housed within body **302**. As noted above, first hoist **306** and second hoist **308** may be embodied as electric reels. Motors for operating the electric reels are not illustrated, as motors may be included within first hoist **306** and second hoist **308**, or may be external to the hoists. First hoist **306** and second hoist **308** can be secured or at least partially secured to body **302**. As illustrated, communication channel **314** provides communication to first hoist **306** and second hoist **308** from a UAV (not illustrated). Communication channel **314** is intended to represent any number of communication channels, and in one example, is intended to represent a channel for providing power supplied from a UAV power source to a first motor associated with first hoist **306** and a second motor associated with a second hoist **308**.

Communication channel **314** may also provide a communication bus from each of first hoist **306** and second hoist **308**, such that each of the hoists can receive instructions from a flight controller and operate in accordance with the instructions, such as to extend or retract a line, and the rate at which to do so.

(48) One example method for operating first hoist **306** and second hoist **308** is to rotationally secure the hoists to body **302**. An example method utilizes cylindrical shaft **310** and cylindrical shaft **312** to rotationally secure first hoist **306** and second hoist **308**, respectively. Using this method, cylindrical shafts may be secured to the hoists. By rotating a cylindrical shaft, the hoist (again, which may include a reel of an electric reel) is rotated, allowing it to extend or release a corresponding line. Here, a motor may engage the cylindrical shaft, such that operation of the motor rotates the cylindrical shaft, thereby rotating the hoist.

(49) Another method that may be used in addition to or in lieu of that just described, includes a stationary securing mechanism that secures the hoist to the body. As examples, brackets, threaded systems, rivets, among many other examples can be used to secure a hoist to a body of a multiple hoist system. FIG. 3, illustrates an example stationary arm **316** that secures first hoist **306** and another example stationary arm **218** that secures second hoist **308** to body **302**. Using stationary securing devices, a motor may be provided as part of a hoist, such that the motor can rotate a portion of the hoists to raise and lower a line. Various methods of securing hoists to a body of a multiple hoist system will be recognized by those having ordinary skill in the art, and a suitable method can be selected based on the choice of hoist being utilized with the multiple hoist system.

(50) Turning now to FIG. 4, a top and side perspective view of example multiple hoist system **400** is provided. Multiple hoist system **400** comprises body **402** having upper surface **403**. One or more securing hooks, such as securing hooks **404A** and **404B**, are affixed to upper surface **403** and can be used to releasably secure multiple hoist system **400** to a UAV for parcel delivery. In another embodiment, multiple hoist system **400** may be permanently secured to the UAV. Body **402** fully

houses first hoist **406** and second hoist **408**. First hoist **406** is associated with first line **412**, while second hoist **408** is associated with second line **414**. First line **412** extends from inside body **402** to an outside of body **402** through first opening **416** on bottom surface **410** of body **402**. Second line **414** is also illustrated as extending from the inside of body **402** to the outside of body **402** through second opening **418** on bottom surface **410** of body **402**.

(51) With reference now to FIG. **8**, a block diagram for an example method **800** for delivering parcels using a multiple hoist system is provided. The method includes, at block **810**, loading a first parcel onto a UAV by securing the first parcel to a first line associated with a first hoist of the UAV and loading a second parcel onto the UAV by securing the second parcel to a second line associated with a second hoist of the UAV, where each hoist is independently operable. This can be performed, in some embodiments, after releasably engaging the securing device of the body with the securing area of the UAV. However, as described above, the body can alternatively be fixed to or integral with the UAV without departing from the scope of the technology herein.

(52) At block **820**, the UAV is navigated to a first delivery location associated with the first parcel. From an altitude above the first delivery location, the method includes, at block **830**, lowering the first parcel downward to the first delivery location using the first hoist and releasing the first parcel at the first delivery location. At block **840**, the UAV is navigated away from the first delivery location to a second delivery location associated with the second parcel. At block **850**, the second parcel is lowered downward to the second delivery location using the second hoist, and the second parcel is released at the second delivery location. At block **860**, the UAV is navigated away from the second delivery location.

(53) In addition to the steps described above, some embodiments of the method **800** can further comprise the flight controller receiving indication that the first parcel or the second parcel has reached the first delivery location or the second delivery location. As a result of this indication, the flight controller can further instruct release of the first parcel or the second parcel from the first line or the second line. For example, a claw mechanism (not shown) may be communicably coupled to the flight controller and actuatable to close or open upon commands therefrom. However, other mechanisms such as hooks or the like can attach to and detach from the first and second parcels without departing from the scope of the technology described herein. Upon release of the first parcel or the second parcel, the flight controller can further instruct the first hoist or the second hoist to retract the corresponding first line or second line via the motor or any other actuators used by the first or second hoists.

(54) With reference to FIG. **9**, an example computing device **900** is provided. Computing device **900** includes bus **910** that directly or indirectly couples the following devices: memory **912**, one or more processors **914**, one or more presentation components **916**, input/output ports **918**, input/output components **920**, and illustrative power supply **922**. Bus **910** represents what may be one or more busses (such as an address bus, data bus, or combination thereof). Although the various blocks of FIG. **9** are shown with lines for the sake of clarity, in reality, delineating various components is not so clear, and metaphorically, the lines would more accurately be grey and fuzzy. For example, one may consider a presentation component such as a display device to be an I/O component. Also, processors have memory. We recognize that such is the nature of the art, and reiterate that the diagram of FIG. **9** is merely illustrates an example computing device that can be used in connection with one or more embodiments of the present technology. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “hand-held device,” etc., as all are contemplated within the scope of FIG. **9** and reference to “computing device.”

(55) Computing device **900** typically includes a variety of computer-readable media. Computer-readable media can be any available media that can be accessed by computing device **900** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable media may comprise computer storage media and communication media.

(56) Computer storage media include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computing device **600**.

Computer storage media excludes signals per se.

(57) Communication media typically embodies computer-readable instructions, data structures, program modules or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

(58) Memory **912** includes computer storage media in the form of volatile or nonvolatile memory. The memory may be removable, non-removable, or a combination thereof. Example hardware devices include solid-state memory, hard drives, optical-disc drives, etc. Computing device **900** includes one or more processors that read data from various entities such as memory **912** or I/O components **920**. Presentation component(s) **916** present data indications to a user or other device. Examples of presentation components include a display device, speaker, printing component, vibrating component, etc.

(59) I/O ports **918** allow computing device **900** to be logically coupled to other devices including I/O components **920**, some of which may be built in. Illustrative components include a microphone, joystick, game pad, satellite dish, scanner, printer, wireless device, etc.

(60) Embodiments described above may be combined with one or more of the specifically described alternatives. In particular, an embodiment that is claimed may contain a reference, in the alternative, to more than one other embodiment. The embodiment that is claimed may specify a further limitation of the subject matter claimed.

(61) The subject matter of the present technology is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed or disclosed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” or “block” might be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly stated.

(62) As used in this disclosure, the word “delivery” is intended to mean both “to drop off” and “to pickup,” unless one of the options is impracticable. For example, a “delivery vehicle” is a vehicle capable of picking up a parcel and dropping off a parcel at a location. Words such as “a” and “an,” unless otherwise indicated to the contrary, include the plural as well as the singular. Thus, for example, the constraint of “a feature” is satisfied where one or more features are present. Also, the term “or” includes the conjunctive, the disjunctive, and both (a or b thus includes either a or b, as well as a and b).

(63) From the foregoing, it will be seen that this technology is one well adapted to attain all the ends and objects described above, including other advantages that are obvious or inherent to the structure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is

within the scope of the claims. Since many possible embodiments of the described technology may be made without departing from the scope, it is to be understood that all matter described herein or illustrated the accompanying drawings is to be interpreted as illustrative and not in a limiting sense.

Claims

1. A multiple hoist system comprising: a body configured to releasably couple to an unmanned aerial vehicle (UAV) at a securing area of the UAV; a first hoist comprising a first reel and a second hoist comprising a second reel attached to the body, wherein the first hoist is operable to independently extend a first line, and the second hoist is operable to independently extend a second line; a first curved contact point integral with or fixed relative to the body, wherein the first curved contact point is configured so that when a first force is applied along the first line due to a first load attached to the first line, the first line contacts the first curved contact point to create tension in the first line between the first curved contact point and the first reel, and at least a portion of the first force is distributed to the body; and a second curved contact point integral with or fixed relative to the body, wherein the second curved contact point is configured so that when a second force is applied along the second line due to a second load attached to the second line, the second line contacts the second curved contact point to create tension in the second line between the second curved contact point and the second reel, and at least a portion of the second force is distributed to the body.
2. The multiple hoist system of claim 1, wherein: the first reel is operable for selectively extending and retracting the first line and the first hoist comprises a first line leveler movable parallel to the first reel for evenly winding the first line, and the second reel is operable for selectively extending and retracting the second line and the second hoist comprises a second line leveler movable parallel to the second reel for evenly winding the second line.
3. The multiple hoist system of claim 1, further comprising a line cutter operable to sever at least one of the first line or the second line.
4. The multiple hoist system of claim 3, wherein the line cutter is operable to sever at least one of the first line or the second line based at least in part on the first force applied to the first line or the second force applied to the second line satisfying a threshold.
5. The multiple hoist system of claim 1, wherein each of the first hoist and the second hoist is selectively operable to synchronously extend the first line and the second line.
6. The multiple hoist system of claim 1, wherein at least one of the first hoist or the second hoist further comprises a line release configured to selectively prevent or oppose a force of gravity from further releasing at least one of the first line or the second line based on altitude.
7. A system comprising: an unmanned aerial vehicle (UAV) having a securing area; and a hoist system comprising: a body configured to couple to the UAV at the securing area, a hoist comprising a reel that is operable to extend a line, and a curved contact point integral with or fixed relative to the body, wherein the curved contact point is configured so that when a force is applied along the line due to a load attached to the line, the line contacts the curved contact point to create tension in the line between the curved contact point and the reel, and at least a portion of the force is distributed to the body.
8. The system of claim 7, wherein the reel is operable for extending and retracting the line and the hoist further comprises a line leveler movable parallel to the reel for evenly winding the line.
9. The system of claim 7, wherein the UAV further comprises a controller and the hoist system further comprises a line cutter operable to sever the line in response to instructions from the controller.
10. The system of claim 9, wherein the controller is operable to send the instructions to the line cutter in response to at least one of an upward thrust of the UAV or the force applied to the line satisfying a threshold.

11. The system of claim 7, wherein the hoist comprises a line release configured to selectively prevent or oppose a force of gravity from further releasing the line based on altitude.
 12. A hoist system comprising: a body configured to couple to an unmanned aerial vehicle (UAV); a first hoist comprising a first reel and a second hoist comprising a second reel attached to the body, wherein the first hoist is operable to independently extend a first line and the second hoist is operable to independently extend a second line; and a curved contact point integral with or fixed relative to the body, wherein the curved contact point is configured so that when a force is applied along the first line due to a load attached to the first line, the first line contacts the curved contact point to create tension in the first line between the curved contact point and the first reel, and at least a portion of the force is distributed to the body.
 13. The hoist system of claim 12, wherein the first reel is operable for extending and retracting the first line and the first hoist further comprises a line leveler movable parallel to the first reel for evenly winding the first line.
 14. The hoist system of claim 12, further comprising a line cutter operable to sever the first line.
 15. The hoist system of claim 14, wherein the line cutter is operable to sever the first line based at least in part on the force applied to the first line satisfying a threshold.
 16. The hoist system of claim 12, wherein the first hoist further comprises a line release configured to selectively prevent or oppose a force of gravity from further releasing the first line based on altitude.
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